M. SIVA NAGA RAJU

Senior Technical Analyst | AI Solutions Architect | Migration Specialist

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Portfolio: https://sivanagarajumolabant.github.io/Portfolio/

SUMMARY

Senior Technical Analyst with **7+ years of experience** at Quadrant Technologies, specializing in enterprise-grade AI-powered database migration solutions for the QMigrator product. Architect of 6 production-grade autonomous agents built on LangGraph and LangChain, deployed across Fortune 500 clients including UHG, Elevance Health, eBay, Nike, and Apollo Hospitals.

- Achieved 98% automation in database schema conversion across Oracle, PostgreSQL, SQL Server, Snowflake, and Databricks
- Delivered 95% AI-driven error correction accuracy with minimal human intervention
- Reduced migration error resolution time by 75% — from 2 hours to 8 minutes
- Drove $2M+ cost savings for enterprise clients through end-to-end automation

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SKILLS

AI & Generative AI: LangChain, LangGraph, Multi-Agent Systems, RAG Architecture, Prompt Engineering, Pydantic AI, Agentic AI, NLP, Conversational AI
Large Language Models: Azure OpenAI GPT-4, Anthropic Claude, Google Gemini, Groq, Ollama
Vector Databases: ChromaDB, FAISS, Pinecone, Milvus, Qdrant
AI Evaluation: DeepEval, Ragas, AI Guardrails, AI Safety, Quality Assurance
Embeddings: OpenAI Embeddings, HuggingFace Embeddings, Ollama Embeddings
Programming Languages: Python (Expert), SQL (Advanced), JavaScript
Backend Frameworks: FastAPI, Django, Django REST Framework, Flask, Streamlit
API Development: RESTful APIs, GraphQL, Swagger, OpenAPI
Data Engineering: Pandas, NumPy, Apache Airflow, Celery, ETL Pipelines, Matplotlib, Seaborn
Databases: PostgreSQL, Oracle, SQL Server, MySQL, MongoDB, Redis
Data Platforms: Snowflake, Databricks, Delta Lake
Database Skills: Query Optimization, Performance Tuning, PL/SQL, PL/pgSQL, T-SQL, Index Design
Cloud - Azure: AKS, ACR, Azure DevOps, Key Vault, Pipelines, Azure PostgreSQL

Cloud - AWS: ECS, Fargate, ECR, API Gateway, Lambda, S3, EC2, RDS
DevOps & Containers: Docker, Kubernetes, Helm Charts, GitHub Actions, GitLab CI/CD
Frontend: ReactJS, JavaScript, HTML5, CSS3, Material UI
Testing: Pytest, Playwright, Selenium
Tools: Git, GitHub, GitLab, Bitbucket, JIRA, Confluence, Postman, VS Code, PyCharm, Cursor, Windsurf

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WORK EXPERIENCE

Quadrant Technologies — Hyderabad, India

Senior Technical Analyst | March 2019 - Present (7+ Years)

Product: QMigrator — Enterprise Database Migration Platform

Core Product Developer and AI Solutions Architect | 2019 - Present
Platform URL: https://qcam.qmigrator.ai/

QMigrator is Quadrant Technologies' flagship enterprise database migration platform that automates end-to-end schema conversion, data migration, and code translation across heterogeneous database environments. Key contributor since inception — built and evolved the platform from a feature management system to a full-scale AI-powered migration solution.

- Built and maintained the core migration platform serving 15+ enterprise clients over 7 years
- Evolved the product from manual feature-based conversion (2019) to fully autonomous AI-driven migration (2025)
- Processed 100,000+ database objects across Oracle, PostgreSQL, SQL Server, Snowflake, and Databricks
- Delivered 15+ large-scale enterprise migrations with 98% automation rate

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QMigrator Development: AI Agents Module | 2025 - Present

Designed and deployed 6 production-grade autonomous agents integrated into QMigrator, revolutionizing database migration workflows across Fortune 500 engagements.

Agent 1: Conversion Agent (Stage 1) — Autonomous Migration Error Resolution - Automated 80% of migration error corrections, reducing resolution time from 2 hours to 8 minutes - Processed 50,000+ migration errors across 8 enterprise clients with 95% success rate - Integrated multi-provider LLM support (GPT-4, Claude, Gemini) with dynamic model selection and fallback - Supports Oracle-to-PostgreSQL, SQL Server-to-PostgreSQL, Oracle-to-SQL Server, Snowflake-to-Databricks - Stack: Python, LangChain, LangGraph, FastAPI, PostgreSQL, Oracle, Azure OpenAI

Agent 2: Conversion Agent (Stage 2) — Module Enhancement Pipeline - Improved QMigrator conversion accuracy by 40% through root-cause module fixing - Achieved 90% performance gain via enhanced module registry and smart caching - Eliminated 60% of recurring conversion failures through systematic module updates - Stack: Python, LangGraph, FastAPI, Dynamic Module Loading, Excel Processing

Agent 3: Debug Mode Agent — Human-in-the-Loop Iterative Debugging - Reduced complex error resolution time by 65% through intelligent snippet isolation - Achieved 95% approval rate for AI-suggested fixes with human validation - Processed 10,000+ debugging sessions with complete audit trails and real-time feedback - Stack: Python, LangGraph, FastAPI, Threading, PostgreSQL

Agent 4: DBA Agent — AI-Powered Database Health and Analysis - Automated 70% of routine DBA queries, reducing manual database inspection time - Processed 25,000+ database queries with 90% accuracy via natural language interface - Supports PostgreSQL, Oracle, SQL Server, and MySQL with real-time query execution - Stack: Python, LangChain, LangGraph, FastAPI, psycopg2, cx_Oracle, SQLAlchemy

Agent 5: Performance Agent — SQL Optimization and Tuning - Optimized 15,000+ SQL queries with an average 60% performance improvement - Achieved 90% cache hit rate, reducing query response time from minutes to milliseconds - Includes execution plan analysis, index recommendations, and intelligent query rewriting - Stack: Python, LangGraph, FastAPI, PostgreSQL, Query Plan Analysis, SQLAlchemy

Agent 6: Dev Conversion Agent — Development and Testing Workflow - Accelerated development cycles by 50% through visual workflow monitoring - Reduced debugging time by 40% via interactive state inspection and step-by-step execution - Built real-time visualization UI with parameter tuning and export functionality - Stack: Python, Streamlit, LangGraph, FastAPI, Pandas

QMigrator Development: Multi-LLM Infrastructure | 2025 - Present

- Architected enterprise-grade LLM abstraction layer supporting Azure OpenAI, Anthropic Claude, Groq, Google Gemini, and Ollama
- Built dynamic model switching with cost optimization, fallback mechanisms, and token usage tracking
- Integrated AI guardrails and evaluation frameworks (DeepEval, Ragas) for quality assurance and safety
- Stack: Python, LangChain, DeepEval, Ragas, JSON/YAML Configuration

QMigrator Development: Core Migration Engine | 2021 - 2022

- Developed core migration engine components with modular Python architecture
- Built Django REST Framework APIs and React-based UI for migration workflow management
- Designed extensible plugin architecture supporting 50+ migration feature types

QMigrator Development: QBook V2 — Feature Management Platform | 2020 - 2021

- Created self-service platform enabling developers to register and manage migration features
- Built multi-level migration types, approval workflows, and version control for feature modules
- Developed full-stack Django + React application with admin dashboard serving 30+ developers

QMigrator Development: QBook V1 — Feature Management System | 2019 - 2020

- Established foundational architecture for feature-based migration approach
- Implemented developer registration, authentication, and approval workflows
- Built Python-based feature templates for standardized development across teams

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QMigrator Client Deployments

Project: NIKE — Cloud Data Platform Modernization

Snowflake to Databricks Migration | 2026 - Present

- Leading large-scale data platform migration for enterprise analytics on Databricks
- Designed ETL pipeline transformations achieving 98% data migration accuracy with automated validation
- Optimized query performance with 40% faster execution times on Databricks
- Deployed on AWS infrastructure with Delta Lake architecture for data lakehouse implementation

Project: eBAY — Enterprise Database Migration

Oracle to PostgreSQL Migration | 2023 - Present

- Deployed Stage 1 and Stage 2 conversion agents for automated error correction pipeline
- Implemented Performance Agent for query optimization during migration
- Achieved 98% automation in schema conversion with AI-driven validation
- Deployed on AWS using ECS with Fargate, ECR, and API Gateway

Project: ELEVANCE HEALTH (Formerly Anthem) — Enterprise Migration

SQL Server to Azure PostgreSQL | 75,000+ Objects | 2024 - 2025

- Orchestrated AI agent workflows for complex PL/SQL to PL/pgSQL conversions
- Implemented Debug Mode for human-in-the-loop validation of critical business logic
- Achieved 95% conversion accuracy for stored procedures and functions
- Reduced manual intervention by 80% through intelligent agent automation

Project: UHG (United Health Group) — Enterprise Migration

Oracle to PostgreSQL | 50,000+ Objects, 2TB+ Data | 2023 - 2025

- Deployed all 6 AI agents to production serving 100+ concurrent users
- Achieved 98% schema conversion automation with AI-driven error correction
- Implemented Kubernetes-based architecture for horizontal scaling
- Reduced migration timeline by 60% compared to traditional approaches with zero-downtime deployment

Project: APOLLO HOSPITALS — Healthcare System Migration

On-Premise Oracle to Azure PostgreSQL | 2022 - 2024

- Managed weekly release cycles with comprehensive deployment validations
- Coordinated cross-functional teams (DBA, Application, Infrastructure) for seamless rollouts
- Ensured 100% compliance with healthcare data security standards (HIPAA)
- Achieved zero data loss during mission-critical healthcare system migration

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EDUCATION

Master of Technology (M.Tech) in Computer Science and Engineering
KITS Engineering College, JNTU Kakinada — Kakinada, Andhra Pradesh

Bachelor of Technology (B.Tech) in Computer Science and Engineering
Chalapathi Engineering College, JNTU Kakinada — Guntur, Andhra Pradesh

Diploma in Electrical and Electronics Engineering
Al-Huda Polytechnic College, AICTE — Nellore, Andhra Pradesh

CERTIFICATIONS

- Advanced LangChain and LangGraph Patterns for Production Systems
- Azure Solutions Architect — Cloud Infrastructure and DevOps
- Kubernetes Application Development and Container Orchestration
- Database Performance Tuning and Query Optimization
- Agile and Scrum Methodologies — Sprint Planning and Team Leadership

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LANGUAGES

- English: Professional Working Proficiency
- Telugu: Native
- Hindi: Professional Working Proficiency